Results: At last follow-up, 17 patients in the re-irradiated group survived without metastasis, 1 was alive with metastasis, 10 were dead of metastasis and 3 of other causes. In the enucleated group, 15 patients were alive without disease, one was alive with metastasis, 25 were dead of metastasis and 2 of other causes. Median survival duration in enucleated and re-irradiated groups was, respectively, 42 and 90 months ($p = 0.040$, log-rank test). Median time free of metastases was 38 months in enucleated and 97 months in re-irradiated patients ($p = 0.028$, log rank-test). At 5 years after enucleation or re-irradiation, the probability of overall survival was, respectively, 36% and 63%; the probability of freedom from metastases was, respectively, 31% and 66%. However, the differences in risk of all cause mortality and of onset of metastasis were not significant after adjustment for recurrent tumor volume in the proportional hazards model. Nine eyes had been enucleated in the irradiated group, 5 for local recurrence and 4 because of intractable pain. The probability of survival with eye retention at 5 years was 40% for the re-irradiated group.

Conclusions: This retrospective analysis suggests that survival in re-irradiated patients is not compromised by administration of a second course of PBRT for recurrent uveal melanoma. The study also shows that patients with uveal melanoma recurrent after high dose fractionated proton beam therapy can be re-irradiated with the same modality using the same dose as given initially, with acceptable morbidity, and with a reasonable expectation of eye retention.

1092 Effectiveness of Fractionated Stereotactic Radiation Therapy for Uveal Melanoma

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Purpose/Objective: Uveal melanoma has traditionally been treated by enucleation. Since 1970, various eye sparing treatment modalities has been developed with as aim an optimal sparing of visual function. Brachytherapy appears to be a good alternative for enucleation in the treatment of small uveal melanoma, according to recent comparative studies. For medium sized tumors, proton beam and helium ion radiation techniques are widely used. Radiosurgery for uveal melanomas has been introduced in 1987 with excellent results of local control. The most important disadvantage of single fraction high dose radiosurgery is an enhanced complication probability, which causes the loss of radiobiological sparing if critical structures are adjacent to the tumor. Fractionated stereotactic radiotherapy reduces the effective dose in the surrounding tissues with a delivery of an equivalent effective dose to the tumor.

This study analyses the effectiveness and side effects of fractionated Stereotactic Radiation Therapy (f.SRT) for uveal melanoma in a prospective, nonrandomized, self-controlled clinical trial.

Materials/Methods: Between 1999 and 2003, 38 uveal melanoma patients (21 male, 17 female) with mean tumor thickness of 6.4 mm were included in the study. Informed consent was obtained from all patients. The mean follow-up was twenty-five months, with a minimal follow-up of 10 months. No patients were lost to follow-up. A total dose of 50 Gy, prescribed to the 80% isodose line, was given in 5 daily fractions of 10 Gy by LINAC-based SRT. A blinking light in front of the unaffected eye and a camera to monitor the position of the diseased eye are fixed to a non-invasive relocatable stereotactic frame. The main primary outcome measures were local tumor control and visual acuity at 3, 6, 12 and 24 months respectively and acute toxicity (peri- and intraocular side-effects).

Results: After three months (38 patients) the local control was 100%, after 12 months (32 patients) one patient was treated by transpupillar thermotherapy because the tumor size did not decrease and after 24 months (15 patients) no recurrences were seen. The visual acuity gradually declines from a mean of 0.21 at diagnosis to a mean of 0.06 two years after therapy. The acute side effects after three months were: conjunctival symptoms (10), loss of eye lashes (6), visual symptoms (5), fatigue (5), dry eye (1), cataract (1), and pain (4). One eye was enucleated 2 months post f.SRT.

Conclusions: Preliminary results demonstrate that fractionated SRT is an effective and safe treatment modality for uveal melanoma with excellent local control and mild acute side effects. The follow-up period is too short to discuss meaningfully the risk for long term side effects.

1093 Decreased Incidence of Neovascular Glaucoma by Sparing Anterior Structures of the Eye for Proton Beam Therapy of Ocular Melanoma

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Purpose/Objective: We previously hypothesized that sparing the anterior segment of the eye would reduce the incidence of neovascular glaucoma (NVG) (Daftari, I. K., et al., IJROBP 1997, 39:997–1010; Meecham, W.J., et al., Arch Ophthalmol 1994, 112:197–203). To test this hypothesis, we examined the incidence and contributing factors of NVG in patients treated with proton beam therapy at the Crocker Nuclear Lavoratory faculty in Davis, California using techniques for sparing anterior structures of the eye.

Materials/Methods: Using anterior structure sparing techniques, 346 patients were treated with proton beam therapy between June 1996 and November 2003 and included in the ocular melanoma database of the Tumori Foundation. Twenty-four patients were excluded due to prior surgical resection and 3 patients because they were treated for recurrences after ^{125}I plaque therapy. This resulted in 319 evaluated patients treated with anterior structure sparing techniques. The patients in the study received 56 GyE in 4 fractions. The primary endpoint of this study is the occurrence of NVG. The median follow-up is 24.7 months. The log rank test was used to compare distributions of the occurrence of NVG. The likelihood ratio test (LLRT) from Cox's proportional hazard model was used to compare models to identify independent predictors of outcome.

Results: NVG occurred among 39 of the patients. The probability of remaining free of NVG at 3 years was 89%. The smaller the tumor diameter was, the better the outcome with 3 year estimates of NVG for the quartiles $<$7.5 mm, $>$7.5 mm - 10.5 mm,

>10.5 mm - 12.0 mm vs. >12.0 mm of 96%, 81%, 80% and 77%, respectively ($p = 0.001$). When none of the lens received at least 50% of the dose there was a lower probability of NVG ($p < 0.0001$). The further away the tumor was from the fovea or from the disc the later NVG occurred ($p = 0.003$, $p = 0.006$) with the 3 year estimates for distances of 0 and >0 mm of 72% and 89% to the fovea and 76% and 88% to the disc. If <100% of the macula and optic disc and less than 29% (upper quartile) of the ciliary body received at least 50% of the dose, the NVG outcome was more favorable ($p = 0.02$, $p = 0.006$ and $p < 0.0001$). Factors in univariate analysis that were not predictors of NVG are age of the patient, gender, or the eye involved.

When the factors determined to be significant predictors of NVG with univariate methods are considered simultaneously, the most significant predictor of NVG is when at least 30% of the ciliary body received at least 50% of the dose (LLRT: $p < 0.0001$). There is also a significant association between the macula and optic disc being treated with at least 50% of the dose ($p < 0.0001$). For 50% of the patients <100% of both sites and for 23% of the patients 100% of both sites were treated with at least 50% of the dose. Therefore, using these cut-points, the final predictive model was either ciliary body proportion plus macula or plus disk proportion (LLRT: $p = 0.003$).

Conclusions: We have previously shown that only. 65% of patients treated with particle therapy for ocular melanoma without anterior structure sparing techniques remained free of NVG at three years of follow-up. These data demonstrate that prospectively sparing anterior structures in the eye increases the probability of remaining free of NVG to 89% at three years. Additional new findings are the strong influence of ciliary body dose, and the effect of irradiation of the macula or the disc on NVG risk.

1094 Proton Radiosurgery in the Management of Functioning and Non-Functioning Pituitary Adenomas: A 10-year Experience at the Massachusetts General Hospital

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Purpose/Objective: To evaluate the efficacy and safety of stereotactic proton radiosurgery (P-SRS) delivered using CT/MRI-based treatment planning for functioning (F) and non-functioning (NF) pituitary adenomas.

Materials/Methods: Between August 1993 and July 2003, 61 patients were treated with P-SRS for residual or recurrent pituitary adenomas at Massachusetts General Hospital. Fifty patients presented with biochemical evidence of residual or recurrent Cushing disease ($n = 27$), acromegaly ($n = 18$), prolactinoma ($n = 3$), or TSH-secreting tumor ($n = 2$). Eleven patients had NF adenomas with evidence of local progression. Median age was 41 years (range 14–79). Sixty patients had prior surgery with a median of 2 transspenoidal resections (range 1–4) and 1 patient had none. P-SRS was delivered at the Harvard University Cyclotron (1993–2001) or Northeast Proton Center (2002–2003) using 2–5 convergent beams. Treatment planning was based on fused CT/MRI images in all cases. Median dose was 20 CGE (range 15–24) for F adenomas and 17 CGE (range 15–20) for NF adenomas. Median treatment volume was 1.3 cc (range 0.45–6.3) for F adenomas and 1.6 cc (range 1.1–3.7) for NF adenomas. The dose to the optic chiasm was limited < 8 CGE in all cases. Endocrine follow-up included serial testing of urine free cortisol for Cushing disease, IGF-1 for acromegaly, prolactin for prolactinoma, and free T4 for TSH-secreting tumor. The primary endpoint for F tumors was normalization of endocrine testing without medical suppression (complete response). The secondary endpoint was normalization of endocrine testing on continued medical suppression (partial response). The primary endpoint for NF tumors was local control based on both MRI and clinical follow-up. Patient and treatment characteristics were evaluated for statistical association with outcomes.

Results: Complete endocrine and radiographic follow-up was available for 57 patients (94%) at a median 34 months (range 8–107) after treatment. Complete response was obtained in 12 patients with Cushing disease (48%), 10 with acromegaly (63%), and 2 with prolactinoma (66.6%). Median time to complete response was 20.5 months (range 6–60). Partial response was observed in 7 patients with Cushing disease (28%), 3 with acromegaly (19%), 1 with prolactinoma (33.3%), and 2 with TSH-secreting tumor (100%). Complete or partial response rates were 76%, 81%, 100%, and 100% for Cushing disease, acromegaly, prolactinoma, and TSH-secreting tumor, respectively. All patients with NF adenomas maintained local control and 3 patients (27%) demonstrated greater than 50% tumor shrinkage. Nine patients with NF adenomas presented with visual disturbances that resolved in 3 patients (33%), improved in 4 (44%), and remained stable in 2 (22%). Poor response was associated with Nelson Syndrome (50% vs. 85%, p = NS) and cavernous sinus involvement (67% vs. 90%, $P = 0.04$). Among 49 patients with pituitary function, 9 (18%) had partial pituitary failure and 2 (4%) had complete pituitary failure after treatment. Median time to pituitary failure was 17 months (range 2–41). No visual complications were reported. Radiation effect in the temporal lobe was reported in MRI follow-up for 2 patients and was associated with temporary seizure activity for 6 months in 1 patient. Both patients had been treated for adjacent cavernous sinus involvement and neither sustained long-term injury.

Conclusions: P-SRS is efficacious in the management of residual or recurrent pituitary adenomas. This study supports a body of literature suggesting that radiosurgery results in expeditious biochemical response with minimal morbidity. P-SRS is advantageous for larger tumors and those in close proximity to the optic pathways where other radiosurgical techniques are limited.

	Complete Response	Partial Response	Overall Response
Cushing Disease (N=25)	12 (48%)	7 (28%)	19 (76%)
Acromegaly (N=16)	10 (63%)	3 (19%)	13 (81%)
Prolactinoma (N=3)	2 (67%)	1 (33%)	3 (100%)
TSH-secreting (N=2)	0 (0%)	2 (100%)	2 (100%)